

Project Proposal

Library Management System (LMS)

Group Information

Class: BCS-4D

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1. Problem Statement

Traditional library setups rely heavily on manual record-keeping, which is prone to human error, inefficiency, and data loss. Librarians struggle with tracking physical book circulation, managing member borrowing histories, and accurately calculating overdue fines. For members, the lack of a centralized system creates confusion regarding book availability and due dates. There is a strong need for a digitized, automated platform to streamline these core operations and improve the user experience for both staff and library members.

2. Proposed Solution & Objectives

We propose the development of a modern, web-based Library Management System (LMS). The primary objective of this project is to create a centralized, automated platform that digitizes everyday library operations. Key objectives include:

- Automating the book circulation process to securely link members with borrowed books.
- Implementing a programmatic fine calculation mechanism (e.g., penalizing overdue returns at 10 PKR per day).
- Providing real-time analytics and inventory health tracking via an interactive dashboard.
- Ensuring a seamless, zero-reload user experience using a Single-Page Application (SPA) architecture.

3. Proposed Technology Stack

To ensure performance and maintainability, the system will be built using the following lightweight technology stack:

Frontend: HTML5, CSS3 (Custom properties, responsive layouts), and Vanilla JavaScript (Fetch API/DOM manipulation) for an app-like feel.

Backend: PHP utilizing a RESTful-style API architecture to handle JSON requests and asynchronous operations.

Database: SQLite (via PDO) implementing a normalized schema and complex views, with the capability to scale to PostgreSQL if required.

4. Key Modules to be Developed

- **Dashboard & Analytics:** A central hub displaying real-time metrics, including total books, active members, currently issued books, and aggregated pending fines.
- **Catalogue & Member Management:** Modules allowing librarians to manage library members (tracking privileges and status) and maintain the book catalogue (preventing over-issuance by tracking total vs. available copies).
- **Circulation Module (Issue & Return):** A core module to handle borrowing logic. It will automatically assign a 14-day borrowing period, decrease available inventory upon issue, and restore inventory upon return.
- **Automated Fine Tracker:** A dedicated module to automatically identify overdue returns, calculate the exact penalty amount based on days overdue, and manage payment settlements.

5. Expected Impact

Upon completion, this Library Management System will eliminate the bottleneck of manual record-keeping. By automating inventory tracking and fine calculation, the system will reduce human error, improve operational transparency, and provide a sophisticated, modern tool for efficient library management.